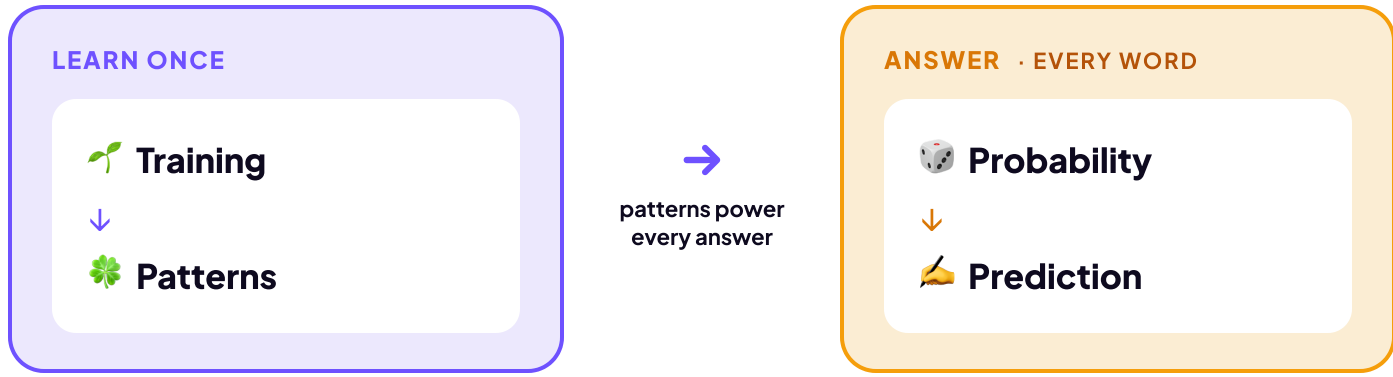


How an LLM Works

You’ve met the two kinds of AI and named the generative one: a Large Language Model, or LLM. So how does an LLM actually turn your words into an answer?

When ChatGPT or Claude write a sentence, they’re running math to predict the likely next words. They aren’t looking up what your words mean; they’re working out which words tend to follow which.

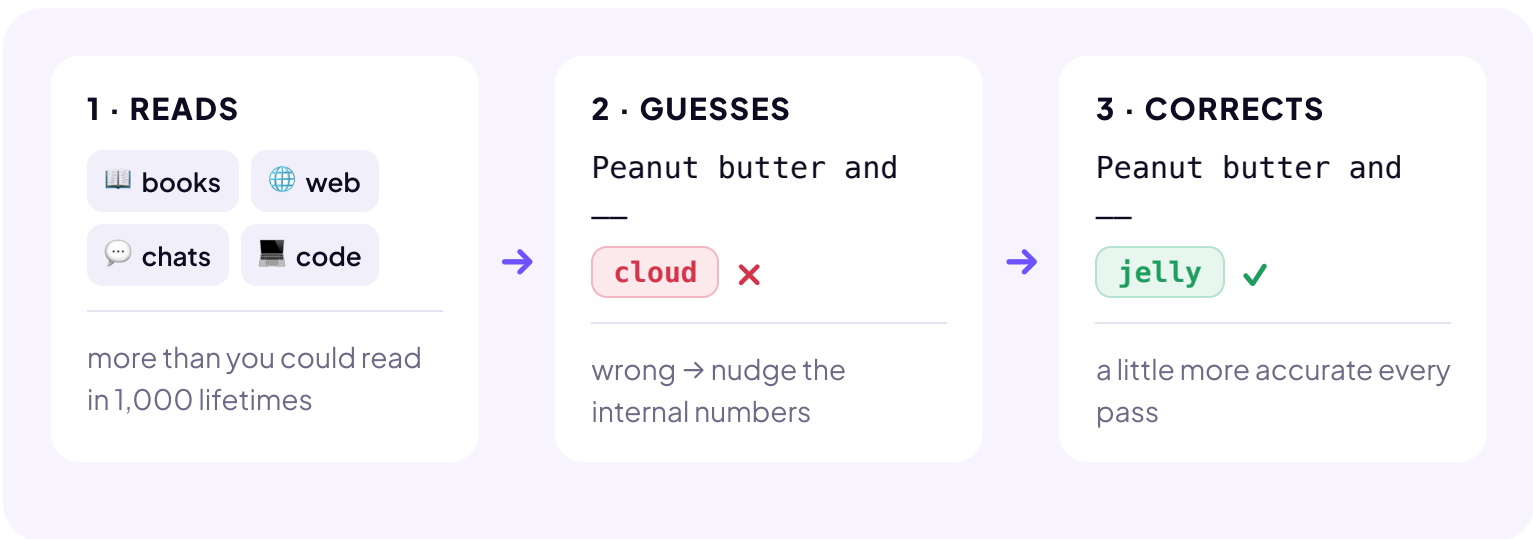
How do they do it? In two phases. First the model learns, once, by soaking up patterns from mountains of text. Then, every time you chat, it uses those patterns to build your answer one word at a time.



Four ideas carry it. Below, we follow one example, peanut butter, through all four.

01 🌱 Training LEARN ONCE

The model **teaches itself**: guess the next word, check, and nudge its numbers toward the right word.



Repeat that loop billions of times, and that's training.

02 🌿 Patterns LEARN ONCE

So what is it actually learning? **Patterns**. Here's one you picked up as a child. Which word comes next?

Peanut butter and ————— → **jelly** *you knew it, so does AI*

PATTERNS ARE EVERYWHERE

Twinkle, twinkle, little — → **star**

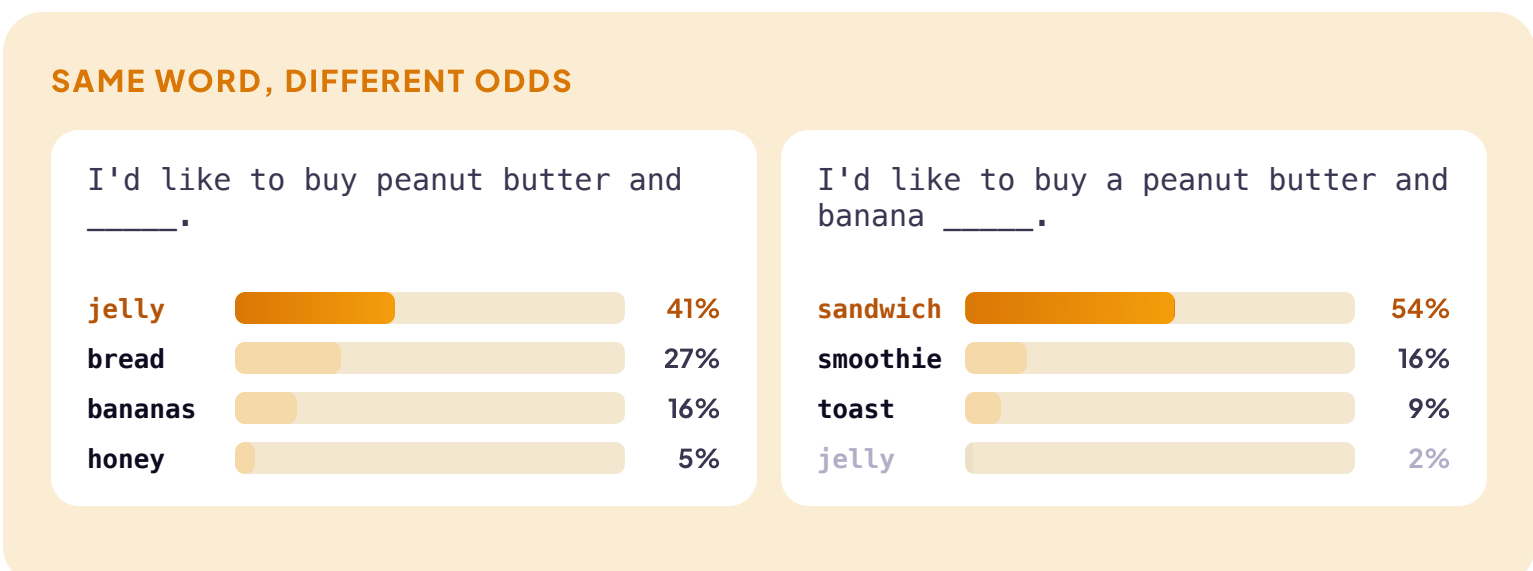
Once upon a — → **time**

Better late than — → **never**

AI didn’t memorize these. It absorbed the pattern by running through billions of examples.

03 🎲 Probability EVERY WORD

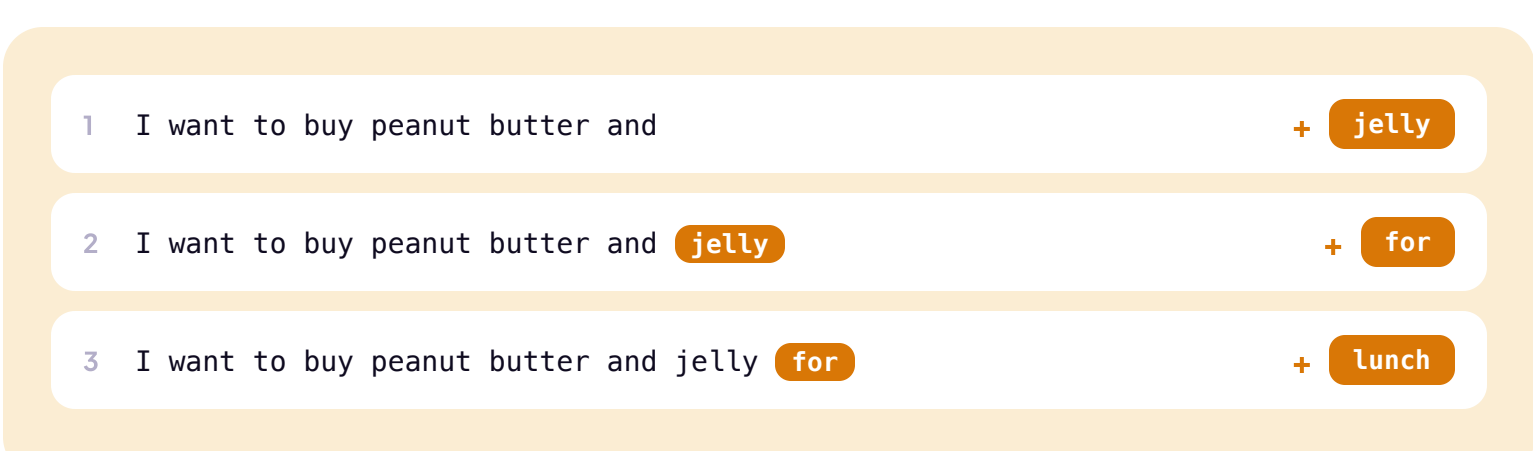
AI doesn’t make one guess. It scores **every** possible next word: a ranked list with a probability on each, and those numbers shift with the surrounding text.



Add the word banana, and jelly’s probability drops from 41% to 2%. The word banana changed everything.

04 🍷 Prediction EVERY WORD

Probability handled one word. But your answer is hundreds of words long, so the model just repeats the move. Your phone does this when you write a text: it suggests a word, you tap it, it suggests the next. AI works the same way, with no fixed plan for where the sentence will end up. One word, look again, the next.



A full paragraph runs this loop many times, fast enough to look like thought.

The infographic 'What is AI?' explains the three core concepts of LLMs. It features a young man holding a smartphone showing a text input. The three steps are: 1. **1 PATTERNS**: AI reads tons of text during training to learn which words tend to follow which. 2. **2 PROBABILITY**: AI looks at the words so far and scores how likely each option is to come next, showing a ranked list for the word 'jelly'. 3. **3 PREDICTION**: AI picks from the top, adds the word, and repeats to build the rest of the answer. A key insight is that AI doesn't 'know' meaning like people do; it finds patterns and predicts the most likely next word. A bottom section titled 'TRAINING' shows AI learning from books, websites, conversations, code, images, audio, and video by reading and watching over and over, updating its internal numbers to get a little better at guessing next words. A final note explains that AI uses what it learned to predict the next word and build an answer, and that different contexts lead to different probabilities, which is why AI might choose 'jelly' or something else!

That clears up three big myths:

- AI isn’t **magic**: it’s math working out probabilities.
- It isn’t **a person**: no thoughts, no understanding, even when it sounds like it has both.
- It isn’t **a truth machine**: it predicts what sounds likely, so a wrong answer can sound just as confident as a right one.

Keep those three straight and much of the confusion falls away.

TRY IT

You just saw what AI really is: math predicting the next word, and not magic, a person, or a truth machine. For each statement, decide whether it’s true or false.